PHYS 210 Electricity and Magnetism

Celebration of Knowledge!

Supplementary Practice 4 - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

Attempt all questions. Supporting practice; this topic mix is not the announced main-test format.

Write all rough calculations and final answers on separate paper. Show your working.

Original source numbering is retained inside each question. Answer the entire cropped question.

---

1. [____] Textbook 15e - Question VP24.11.2

VP24.11.2 The two plates of a parallel-plate capacitor each have area 1.20 m^2, are 4.50 mm apart, and initially have vacuum between them. You connect a power supply to the capacitor, charging it to 3.50 kV. Keeping the power supply connected, you insert a sheet of dielectric with dielectric constant 2.50 that fills the space between the plates.

Find (a) the original capacitance, (b) the original charge on the capacitor plates, (c) the new capacitance once you insert the dielectric, (d) the new charge on the capacitor plates, and (e) the magnitude of the induced charge on the faces of the dielectric sheet.

2. [____] Textbook 15e - Question VP24.11.3

VP24.11.3 With vacuum between its plates, a parallel-plate capacitor has capacitance 4.50 μF. You attach a power supply to the capacitor, charging it to 2.80 kV, and then disconnect it. You then insert a dielectric sheet that fills the space between the plates. The potential difference between the plates decreases to 1.20 kV, and the charge on each plate remains constant. Find the energy stored in the capacitor (a) before you insert the sheet and (b) after you insert the sheet.

3. [____] Textbook 15e - Question VP24.11.4

VP24.11.4 With vacuum between its plates, a parallel-plate capacitor has capacitance 5.60 μF. You attach a power supply to the capacitor, charging it to 1.50 kV. Keeping the power supply connected, you insert a dielectric sheet with dielectric constant 2.85 that fills the space between the plates. Find the energy stored in the capacitor (a) before the sheet is inserted and (b) after the sheet is inserted.

4. [____] Homework 4 - Question 22.15

22.15 •• How many excess electrons must be added to an isolated spherical conductor 26.0 cm in diameter to produce an electric field of magnitude 1150 N/C just outside the surface?